Notation:
- $I^c = [\ell]\setminus I$ for $I \subseteq [\ell]$
- $A^{I,J} = \{\alpha_{ij} \in A : i \in I,\ j \in J\}$ for $A \subseteq R(GL_\ell) = \{\alpha_{ij} : 1 \le i \ne j \le \ell\}$, $I, J \subseteq [\ell]$.
- $\nu_I = (\nu_{i_1}, \dots, \nu_{i_k})$ where $I = \{i_1 < \cdots < i_k\}$, $\nu = (\nu_1, \dots, \nu_\ell)$.
- $\sum A = \sum_{\alpha \in A} \alpha$ = sum of the roots in $A$.

Def: Let $(m,n) \in \mathbb{Z}_+ \times \mathbb{Z}$ be a pair of coprime integers. A Catalanimal $H(R_g, R_t, R_{gt}, \lambda)$ of length $\ell$ is $(m,n)$-cuddly if

(A) it is tame (i.e. $[R_g, R_t] \subseteq R_{gt}$)

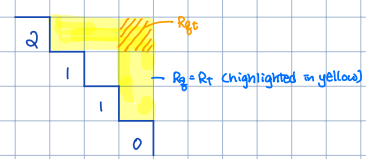
(B) $|\lambda| = \frac{\ell n}{m}$ (hence $m \mid \ell$)

(C) it satisfies the cuddliness bounds $|\lambda[I]_I| \le |I| \cdot \frac{n}{m}$ $\forall\ I \subseteq [\ell]$, where $\lambda[I] = \lambda + \sum R_t^{I,I^c} - \sum R_g^{I,I^c} - \sum R_t^{I,I^c} + \sum R_{gt}^{I,I^c}$.

e.g. $\ell = 4$, $(m,n) = (1,1)$. $R_g = R_t = \{\alpha_{12}, \alpha_{13}, \alpha_{14}, \alpha_{24}, \alpha_{34}\}$, $R_{gt} = \{\alpha_{14}\}$, $\lambda = (2,1,1,0)$.

$\therefore [R_g, R_t] = \{\alpha_{14}\} = R_{gt}$ ($\because H(R_g, R_t, R_{gt}, \lambda)$ is tame.) ; $|\lambda| = 2+1+1 = 4 = \ell = \frac{\ell n}{m}$ ;

$\alpha_{12} + \alpha_{24} = \alpha_{13} + \alpha_{34}$




To check cuddliness bounds:

When $|I| = 0$ or $4$ then the bound trivially holds.

$|\lambda[I]_I| = 0$ ; $|\lambda[I]_I| = |\lambda| = \ell = 4$

When $|I| = 1$:

- $I = \{1\}$.

$$\lambda[I] = (2,1,1,0) + \sum R_t^{\{1\},\{2,3,4\}} - \sum R_g^{\{1\},\{2,3,4\}} - \sum R_t^{\{1\},\{2,3,4\}} + \sum R_{gt}^{\{1\},\{2,3,4\}}$$
$$= (2,1,1,0) + (\alpha_{12}+\alpha_{13}+\alpha_{14}) - (\alpha_{12}+\alpha_{12}+\alpha_{14}) \times 2 + \alpha_{14}$$
$$= (2,1,1,0) - (3,-1,-1,-1) + (1,0,0,-1)$$
$$= (0,0,0,0)$$

$\lambda[I]_I$

- $I = \{2\}$.

$$\lambda[I] = (2,1,1,0) + \sum R_t^{\{2\},\{1,3,4\}} - \sum R_g^{\{2\},\{1,3,4\}} - \sum R_t^{\{2\},\{1,3,4\}} + \sum R_{gt}^{\{2\},\{1,3,4\}}$$
$$= (2,1,1,0) + (\alpha_{23}+\alpha_{24}) - \alpha_{24} - \alpha_{24} + \vec{0}$$
$$= (2,1,1,0) + (0,1,-1,0) - (0,1,0,-1)$$
$$= (2,1,0,1)$$

$\lambda[I]_I$

- $I = \{3\}$.

$$\lambda[I] = (2,1,1,0) + \alpha_{34} - \alpha_{34} - \alpha_{34} = (2,1,0,1)$$

$\lambda[I]_I$

- $I = \{4\}$.

$$\lambda[I] = (2,1,1,0)$$

$\lambda[I]_I$

$\therefore$

| $I$ | $\{1\}$ | $\{2\}$ | $\{3\}$ | $\{4\}$ | |
|---|---|---|---|---|---|
| $\lambda[I]_I$ | $(0)$ | $(1)$ | $(1)$ | $(0)$ | |
| $\lvert\lambda[I]_I\rvert$ | 0 | 1 | 1 | 0 | $\le \lvert I\rvert = 1$ |

When $|I| = 2$

- $I = \{1,2\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{13}+\alpha_{14}+\alpha_{23}+\alpha_{24}) - (\alpha_{13}+\alpha_{14}-\alpha_{24}) \times 2 + \alpha_{14}$$
$$= (2,1,1,0) - \alpha_{13} + \alpha_{23} - \alpha_{24} = (2,1,1,0) + (-1,0,0,1) = (1,1,1,1)$$

- $I = \{1,3\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{12}+\alpha_{14}+\alpha_{34}) - (\alpha_{12}+\alpha_{14}+\alpha_{34}) \times 2 + \alpha_{14}$$
$$= (2,1,1,0) - \alpha_{12} - \alpha_{34} = (1,2,0,1)$$

- $I = \{1,4\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{12}+\alpha_{13}) - (\alpha_{12}+\alpha_{13}) \times 2 = (0,2,2,0)$$

- $I = \{2,3\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{24} + \alpha_{34}) - (\alpha_{24} + \alpha_{34}) \times 2 = (2,0,0,2)$$

- $I = \{2,4\}$

$$\lambda[I] = (2,1,1,0) + \alpha_{23} = (2,2,0,0)$$

- $I = \{3,4\}$

$$\lambda[I] = (2,1,1,0)$$

$\therefore$

| $I$ | $\{1,2\}$ | $\{1,3\}$ | $\{1,4\}$ | $\{2,3\}$ | $\{2,4\}$ | $\{3,4\}$ | |
|---|---|---|---|---|---|---|---|
| $\lvert\lambda[I]_I\rvert$ | 2 | 1 | 0 | 0 | 2 | 1 | $\leq \lvert I\rvert = 2$ |

When $|I| = 3$:

- $I = \{1,2,3\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{14} + \alpha_{24} + \alpha_{34}) - (\alpha_{14} + \alpha_{24} + \alpha_{34}) \times 2 + \alpha_{14}$$
$$= (2,1,1,0) - (\alpha_{24} + \alpha_{34}) = (2,0,0,2)$$

- $I = \{1,2,4\}$

$$\lambda[I] = (2,1,1,0) + (\alpha_{13} + \alpha_{23}) - (\alpha_{13}) \times 2 = (2,1,1,0) - \alpha_{13} + \alpha_{23} = (1,2,1,0)$$

- $I = \{1,3,4\}$

$$\lambda[I] = (2,1,1,0) + \alpha_{12} - \alpha_{12} \times 2 = (1,2,1,0)$$

- $I = \{2,3,4\}$

$$\lambda[I] = (2,1,1,0)$$

$\therefore$

| $I$ | $\{1,2,3\}$ | $\{1,2,4\}$ | $\{1,3,4\}$ | $\{2,3,4\}$ | |
|---|---|---|---|---|---|
| $\lvert\lambda[I]_I\rvert$ | 2 | 3 | 2 | 2 | $\leq \lvert I\rvert = 3$ |

$\therefore$ $H(R_g, R_t, R_{gt}, (2,1,1,0))$ is an $(1,1)$-cuddly Catalanimal.

We will show that $H$ is $(m,n)$-cuddly implies $\psi_p(H) \in \Lambda(X^{m,n})$ (and hence $\mathrm{cub}(H)$ exists).

For this we need the following result from Negut:

**Theorem 4.2.3:**

Let $\mathcal{E}^+_{sp}$ be the subalgebra of $\mathcal{E}^+$ generated by $\Lambda(X^{m,n})$ for $\frac{n}{m} \leq p$.

Given $g = g(z_1, \dots, z_\ell) \in \mathcal{S}^{\mathbf{d}}_{\mathbb{K}}$, then $\psi_p(g) \in (\mathcal{E}^+)^{(\ell,\ell)}$ belongs to $\mathcal{E}^+_{sp}$ iff $g$ is supported on monomials in the $S_\ell$ orbits of dominant weights $\mu$ satisfying $\mu_1 + \dots + \mu_k \leq 2k(\ell-k) + kp \ \forall k \in [\ell]$.

**Prop 4.2.4:** Let $H = H(R_g, R_t, R_{gt}, \lambda)$ be a Catalanimal of length $\ell$ and $g(R_g, R_t, R_{gt}, \lambda) = \sum_{\omega \in S_\ell} \omega\Big(z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R \setminus R_g} (1 - g z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_{gt}} (1 - g t z^\alpha)\Big) \in \mathbb{K}[z_1^{\pm 1}, \dots, z_\ell^{\pm 1}]^{S_\ell}$.

(i) If $H$ satisfies the cuddliness bounds for coprime pair $(m,n) \in \mathbb{Z} \times \mathbb{Z}$, then $g(z)$ satisfies the condition in Theorem 4.2.3 for $p = \frac{n}{m}$.

(ii) If $H$ is $(m,n)$-cuddly, then $H \in \mathcal{S}_p$ with $\psi_p(H) \in \Lambda(X^{m,n})$, so $H$ has a cub.

Proof: For (i), we want to show that for every monomial $z^\nu$ occuring in $g(z)$ and every $I \subseteq [\ell]$ with size $|I| = k$, $|\nu_I| \leq 2k(\ell-k) + k\frac{n}{m}$. If we can show that holds for each term in

$$z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R \setminus R_g} (1 - g z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_{gt}} (1 - g t z^\alpha)$$

then result follows by symmetry.

To get a term $z^\nu$ maximizing $|\nu_I|$ in this product, we must choose $z^\alpha$ term from the factors with $\alpha \in R^{I,I^c}$ (then we add 1 in positions in $I$, hence maximizing $|\nu_I|$) and constant term 1 from the factor with $\alpha \in R^{I^c,I}$.

Take this if $\alpha \in R^{I^c,I}$ Take this if $\alpha \in R^{I,I^c}$

i.e. $1 - * z^\alpha$ (either $1, g, t$ or $gt$)

* For $\alpha \in R^{I,I} \cup R^{I^c,I^c}$, we can take either term b/c $|\alpha_I| = 0$.

Since $|R^{I,I^c}| = |I| \cdot |I^c| = k(\ell-k)$, $|(R \setminus R_g)^{I,I^c}| = k(\ell-k) - |R_g^{I,I^c}|$ and $|(R \setminus R_t)^{I,I^c}| = k(\ell-k) - |R_t^{I,I^c}|$.

Hence
$$|\nu_I| = \Big|\lambda + \sum (R_+ \cap R^{I,I^c}) + \sum ((R \setminus R_g) \cap R^{I,I^c}) + \sum ((R \setminus R_t) \cap R^{I,I^c}) + \sum (R_{gt} \cap R^{I,I^c})\Big|$$
$$= |\lambda| + |R_+^{I,I^c}| + |(R \setminus R_g)^{I,I^c}| + |(R \setminus R_t)^{I,I^c}| + |R_{gt}^{I,I^c}|$$
$$= |\lambda| + 2k(\ell-k) - |R_g^{I,I^c}| - |R_t^{I,I^c}| + |R_{gt}^{I,I^c}| = |\lambda[I]_I| + 2k(\ell-k) \leq 2k(\ell-k) + |I| \cdot \frac{n}{m} = 2k(\ell-k) + k \cdot \frac{n}{m}.$$

by (c) of cuddliness

($\therefore$ (i) follows.)

For (ii), by Prop. 4.1.3, since $H$ is tame, $H \in S_{\hat{P}}$ and from (i), $\mathcal{V}_{\hat{P}}(H) \in \mathcal{E}^{+}_{\leq \frac{n}{m}}$.

$(\mathcal{E}^{+}_{\leq \frac{n}{m}})^{(\ell, \ell \frac{n}{m})} = (\mathcal{E}^{+}_{\leq \frac{n}{m}})^{(\frac{\ell}{m} \cdot m, \frac{\ell}{m} \cdot n)} \subseteq \Lambda(X^{m,n})$ generated by $f(X^{m,n})$ with $\deg f = \frac{\ell}{m}$.

$\because$ $H$ is homogeneous of degree $|\lambda| = \frac{\ell}{m} \cdot n$

$\therefore$ $H$ belongs to $S_{\hat{P}}^{(\ell, \frac{\ell}{m} \cdot n)}$.

By Prop. 3.3.1, $\mathcal{V}_{\hat{P}}(H)$ belongs to $(\mathcal{E}^{+})^{(\ell, \frac{\ell}{m} \cdot n)} \subseteq \Lambda(X^{m,n})$, i.e. $\mathcal{V}_{\hat{P}}(H) \in \Lambda(X^{m,n})$. $\square$